Industrial policy and jobs:

The role of production networks and labor-materiasl substitution

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Industrial Policy and Jobs

- Industrial policy $\longrightarrow$ Particular focus on job creation + distorted economies
  (Grunau, Hoffmann, Lemieux & Mirko Titze, 2025; Martin, Netaraj & Harrison, 2017)

- Effects propagate through IO linkages across sectors (Liu, 2019)

  This paper: Pro-competition employment effect and downstream technology

  - Cheaper materials $\uparrow$ labor demand where complements, but $\downarrow$ labor where substitutes

1. IO linkages $\longrightarrow$ Where the shock travels
2. **Het. labor–materials substitution $\longrightarrow$ Employment effects when the shock arrives**
3. Variable markups $\longrightarrow$ Direct competition effects

Labor–materials complementarity shapes the employment response

For policy, sector choice matters beyond the I/O structure: employment effects depend on the technology of the sectors reached.

Affected sector

- Lower markups and distortions
- Expansion in output and employment
- Higher labor demand through the **scale effect** + **lower distortions**

Downstream sectors (Indirect effect)

- Lower prices of intermediate inputs
- Labor demand depends on labor-material substitution

  Substitutes: ↓ Demand

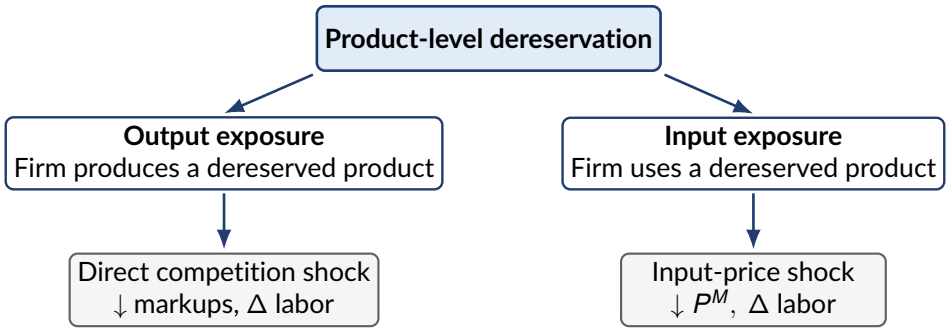
  Complements: ↑ Demand

Reservation and Dereservation Policy in India

- **The Small Scale Reservation Policy (1967-1997):**
  - "Protect" small firms by granting them exclusivity in the production of certain products.
  - Small-Firm Def: Plan & Capital value $\leq$ ₹10'000,000 ($200,000 in 2002)
  - Large firms with export obligations $\geq$ 75% were allowed to produce reserved goods.

- **The Small Scale Dereservation Policy (1997-2015):**
  - Removed production exclusivity restrictions $\longrightarrow$ No capital cap.
  - Lift institutional entry barriers to large firms.
  - Targets around 6% of manuf. sector sales and 10% of employment,

Dereservation generates two distinct shocks



What I do

- **Estimate** labor–materials substitutability sector by sector
- **Measure** the two shocks in the data
  - Staggered product dereservation. ASI plant–product panel 2001–2008
  - Output-exposure and input-exposure samples, held separate
  - Employment effects + Interaction with substitutability.
- **Quantitative** I/O model of Indian manufacturing
  - Variable markups, and a binding **capital cap** on reserved products
- Pro-Competition policy: **Remove** the cap $\longrightarrow$ decompose employment response
  - **Scale** + **Markup** + **Complementarity**

Preview of Results

- Per 100 jobs created in deregulated lines, **57** survive economy-wide.

  Reallocation destroys 60 jobs in never-reserved sectors; complementarity gives back **17**.

- Complementarity mechanism delivers about **one third** of the employment gain (35%).

  83% of the complementarity effect operates through never affected sectors.

- Ignoring complementarities leads to a 30% underestimation of the aggregate effect.

- **Heterogeneity:** a common η assigns about 23 of every 100 jobs to the wrong sectors.

  Two comparable reforms: the one reaching complements creates **44%** more jobs.

The **complementarity** channel is stronger than the classic pro-competition channel (35% vs 18% operating through markups)

Contribution

Industrial policy effects through networks + Labor-Materials complementarities

(Liu, 2019 ; Bonilla, Muñoz-Morales & Zarate, 2026 Atalay, 2017; Castro-Vicenzi & Kleinmann, 2026; Peter & Ruane, 2026; Martin, Netaraj & Harrison, 2017; Garcia-Santana & Pijoan-Mas, 2012)

- Granular sector-level heterogeneity in labor-material elasticity of substitution +
  - Endogenous markups and network effects.
  - Using directly in a structural model replicating the heterogeneity
- Empirical isolation of input and output mechanisms.
- Counterfactual isolation of markups vs complementarity mechanism
  - Distortion vs Technology

Data and Empirical Findings

Annual Survey of Industries (2001-2008)

- Indian Panel of Manufacturing Establishments.
  - Around 114,000 unique firms
- Data on output **and** input prices and quantities
  - Around 5000 different products observed (500 dereserved)
  - Ex. Products: Cotton Bleached Yarn, Black Tea Leaf, Polished Diamonds, Rough Diamonds
- The granularity of the data allows to:
  - Identify firms affected only **directly** through products manufactured
  - Identify firms affected only **indirectly** through inputs consumed

Labor–material complementarity

Firm-level input prices

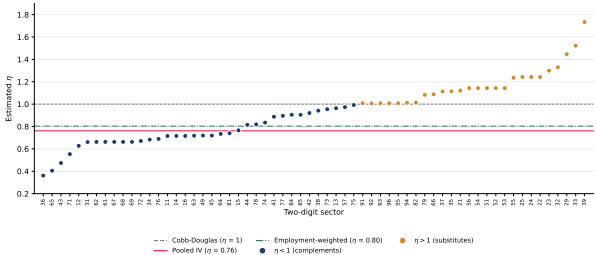
- Use input-level price and quantity data
- Törnqvist input price index, P_{it}^M
- Firm-level wages, w_{it}

Elasticity IV estimation

IV

$$\log\left(\frac{P_{it}^M M_{it}}{w_{it} L_{it}}\right) = \alpha + (\eta_s - 1)\log\left(\frac{w_{it}}{P_{it}^M}\right) + \mathbf{X}_{it}'\gamma + \delta_t + \varepsilon_{it}.$$

- Sectoral (2-digit) η estimates
- $\hat{\eta} < 1 \rightarrow$ **Complements**



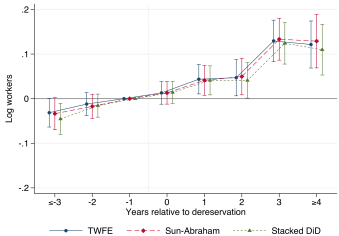
- High heterogeneity: 36 complements, 26 substitutes
- Elasticities $\hat{\eta}$ range from 0.36 to 1.73
- Pooled estimate $\hat{\eta} = 0.76$

Complementarity shapes labor response to industrial policy

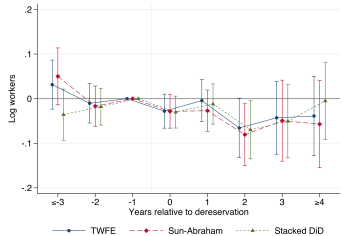
Event-study specification

$$\log L_{it} = \sum_{k \neq -1} \beta_k^C \mathbf{1}\{\tau_{it} = k\} D_s^C + \sum_{k \neq -1} \beta_k^S \mathbf{1}\{\tau_{it} = k\} D_s^S + \alpha_i + \delta_t + \varepsilon_{it}$$

Complements ($\eta_{ls} < 1$)



Substitutes ($\eta_{ls} > 1$)



TWFE, Sun–Abraham and stacked DiD; 95% confidence intervals.

$$D_s^C = \mathbf{1}\{\widehat{\eta}_{ls} < 1\}, \qquad D_s^S = \mathbf{1}\{\widehat{\eta}_{ls} > 1\}$$

τ_{it}: years relative to input dereservation

- Treatment $\rightarrow$ Firms affected through inputs, excluding those affected through outputs.
- Input margin of the policy creates jobs in industries where Labor and Mat. are complements

Industrial Policy Affects Directly Competition and Labor

- Proxy firm-level competition effects using product-level markups.
  (Estimates using De Loecker, Goldberg, Khandelwal and Pavcnik, 2016)
- Treatment $\longrightarrow$ Firms affected through outputs, excluding affected through inputs.
- Output affected firms $\longrightarrow \downarrow$ Markups and $\uparrow$ Labor Demand Markup and Labor
- Labor effects consistent with reduced market distortions.

Model

Model

1. Variable markups depend on endogenous market shares.

2. Restriction in capital stock: Reservation threshold

3. Production network: 3-digit input–output structure.
   Examples of groups: Bakery Products, Leather Apparel

4. CES production: labour and materials with sector-specific η_s.
   - Cobb-Douglas between Labor-Materials CES, and Capital.

5. Entry margin: a predetermined pool of potential entrants.

Analysis Unit in the Model

- Calibrate the model $\longrightarrow$ 3-digit product-groups.
- Within each product group: Policy Affected (R) and Never affected (U) subsectors.
- In R sectors we have a pool of Incumbents (I) and Entrants (E).
- R sectors $\longrightarrow$ Capital cap that introduces distortions:
  - A set of potential entrants $\longrightarrow$ Lower profits than in undistorted market.
  - Binding Incumbents $\longrightarrow$ Less than optimal K and Q
- Industrial policy $\rightarrow$ Cap removal $\rightarrow \uparrow$ Entry and $\uparrow$ Expansion

Model Details: Competition

Final-good aggregator

$$Y = \left(\sum_{s=1}^{S} \Gamma_s^{(1/\theta_f)} C_s^{\frac{\theta_f - 1}{\theta_f}} \right)^{\frac{\theta_f}{\theta_f - 1}}$$

Sectoral aggregator

$$Y_s = \left(\sum_{i \in \mathcal{N}_s} (Y_i^s)^{\frac{\gamma - 1}{\gamma}} \right)^{\frac{\gamma}{\gamma - 1}}$$

Production Function

$$Y_i^s = z_i^s \left[\alpha_t^s L_i^{\frac{\eta_s - 1}{\eta_s}} + (1 - \alpha_t^s) M_i^{\frac{\eta_s - 1}{\eta_s}} \right]^{\beta \frac{\eta_s}{\eta_s - 1}} K_i^{1-\beta}$$

Network: Materials bundle

$$M_i^s = \left[\sum_{s'} \underbrace{\omega^{s,s'}}_{\text{I/O weights}} {M_i^{s'}}^{\frac{\kappa-1}{\kappa}} \right]^{\frac{\kappa}{\kappa-1}}, \quad P^{M,s} = \left[\sum_{s'} \omega^{s,s'\,\kappa} {P^{s'}}^{1-\kappa} \right]^{\frac{1}{1-\kappa}},$$

- κ is the elasticity of substitution across upstream sectors.
- For production, firms in sector s combine inputs from other sectors s'
- Assumption: Within sectors, firms use the same input recipe.

Operational and Sunk Costs

- Policy affected markets and never affected markets incumbents $\longrightarrow$

  Pay f^R and f^U to operate (In wage terms).

- Incumbent profits:

$$\Pi_i^I = p_i^s y_i^s \left(1 - \frac{1}{\mu_i}\right) - f^R$$

- Policy affected markets entrants $\longrightarrow$ Pay f^R and a sunk cost f^E

- Entrant profits:

$$\Pi_i^E = p_i^s y_i^s \left(1 - \frac{1}{\mu_i}\right) - f^R - f^E$$

The capital restriction distorts both scale and entry

- Firms in reserved markets solve their problem subject to

$$K_i \leq \bar{K}.$$

Incumbent scale

$K_i^*(z_i) \rightarrow$ opt. capital without restriction.

$$K_i^R(z_i) = \min\left\{K_i^*(z_i), \overline{K}\right\}.$$

For firms for which the cap binds,

$$K_i^*(z_i) > \overline{K}$$

and

$$K_i^R < K_i^*.$$

Entry

Potential entrant i enters when

$$\Pi_i^E(z_i; \bar{K}) \geq 0,$$

where

$$\Pi_i^E = \pi_i^{var}(z_i; \bar{K}) - f^R - f^E.$$

Define the entry cutoff by

$$\Pi^E(z^E; \bar{K}) = 0.$$

Removing the cap increases maximized variable profits for firms constrained by $\bar{K}$

Complementarity and Employment

- Decompose the labor variation upon market structure changes as follows
  - Labor responses to increases in number of competitors
    Peters, (2022) ; Edmond, Midrigan and Xu, (2015)
  - Labor responses to Δ in competitors in upstream sectors.
    The direction depend on the elasticity of substitution η_s

Illustration with Homogeneous Firms:

$$\hat{L}_i^s = \underbrace{\widehat{\mathrm{Rev}}_s}_{\text{Scale}} \underbrace{- (\eta_s - 1) s_s^m \left[\Omega \Psi \left(\tfrac{1}{\gamma - 1} I + \Gamma\right) \hat{\mathbf{N}}\right]_s}_{\text{Network substitution}} + \underbrace{\Gamma_s \hat{N}_s}_{\text{Competition}} .$$

Calibration

Calibration Is Sector Specific: Many products in a sector

Parameters

- Each 3-digit affected sector s contains:

Reserved branch R and Never-Reserved branch U.

- Sector-specific objects:

η_s, $Input - Output$, market size, firm distribution.

- The **Never-Reserved branch** disciplines the undistorted equilibrium within sector s.
- The **Reserved branch** is subject to

$$K_i \leq \overline{K}.$$

- Remaining parameters reproduce pre-policy differences between the two branches.

Calibration: reproduces the pre-dereservation equilibrium.

Costs

Directly from data

- Input–output shares

$$\omega(s, s')$$

- Capital restriction

$$\bar{K}$$

- Sector size and expenditure shares

Estimated externally

- Labor–material elasticity

$$\eta_s$$

- Demand elasticity

$$\gamma = 6 \quad ; \quad \theta_f = 1.25$$

- Input Substitution

$$\kappa = 2.5$$

Internally calibrated

- Demand / competition parameters
- Operating costs

$$f^R, \ f^U$$

- Entry (sunk) cost

$$f^E$$

- Productivity parameters

Assumptions

Sector-level Pareto productivity distribution + Perfectly elastic labor supply

Disciplining the Model: Entry and Expansion

- **Entry**: Observed entrant count; Pareto productivity anchored to incumbents. Details

- **Productivity**: Disciplined by market share distribution within sectors and HHI.

- **Demand and Capital shifters**: Reserved sectors relative size and capital units.

- **Sunk and Operational Costs**: Operating costs are disciplined by incumbents; sunk costs by entrants' non-entry under the cap and entry after its removal. Details

Calibration Results

Calibration Fit – Concentration Moments

Untargeted

Market	Moment	Sectors	Data	Model	Pearson
Reserved	HHI	79	0.151	0.139	0.94
	Market share p25 (%)	79	0.29	0.28	0.89
	Market share p50 (%)	79	0.96	0.98	0.99
	Market share p75 (%)	79	2.71	2.62	0.97
Never Reserved	HHI	207	0.120	0.167	0.87
	Market share p25 (%)	207	0.16	0.28	0.92
	Market share p50 (%)	207	0.67	0.64	0.76
	Market share p75 (%)	207	1.85	1.70	0.93

- The model reproduces the level and the cross-sector distribution of concentration.

Calibration Fit - Entry and Capital Moments

DiD Expansion

Targeted: post-reform entrants

Moment	Sectors	Data	Model
Entrants per sector	84	12.61	12.69
Entrant sales share (%)	84	19.88	18.48
Entrant P&M p50 / cap	83	0.29	0.39
Entrants above the cap (%)	83	33.1	35.4

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Untargeted: incumbents with Rs 8–10m before the reform

	Data	**Model**
Δ log P&M	0.178	0.071
relative to incumbents below Rs 8m	0.200	0.208
Share above Rs 10m after the reform	0.558	0.613

Counterfactuals

Dereservation Policy: Removal of $\overline{K}$

Two expected effects:

1. Expansion by capped firms close to $\overline{K} \longrightarrow$ Higher output, lower prices. Expansion Validation

2. Re-optimization by entrants that can use now efficiently
   - Lower output and materials prices due to increased competition Effects on Prices
   - 1,210 entrants operate in 104 of the 113 reserved sectors Entry Validation
   - Capped vs Uncapped profits Entrants Profits

Decomposition of the Industrial Policy Effects on Employment: Within group effect per 100 jobs created in deregulated lines

Within %

	Scale	Markup	Complementarity	Total
Deregulated				
$\eta < 1$	+69.1	+8.1	+3.6	+80.8
$\eta > 1$	+17.4	+1.9	−0.2	+19.2
All	+86.5	+10.1	+3.4	+100.0
Never-reserved				
$\eta < 1$	−46.1	0.0	+19.2	−26.9
$\eta > 1$	−13.7	0.0	−2.5	−16.2
All	−59.8	0.0	+16.7	−43.1
Net	+26.7	+10.1	+20.1	+56.9

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- In directly affected sectors, employment gains are **dominated by scale effects**
- The **markup channel** is relevant, but smaller than the **scale effect**

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- Never-reserved lines shed **60** → complementarity gives back **17**
- Complementarity adds **20** of the **57** net jobs → **35%** of the net effect

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- **Complementarity** > **Markup** channel

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- Never-reserved lines shed **60** → complementarity gives back **17**
- Complementarity → **35%** of the net effect
- Driven by the technology distribution in the economy
- Complementarity > Markup channel
- **83%** of the complementarity contribution arises outside the deregulated sectors

Experiments: Complementarity Component per 100 jobs created in deregulated lines

By Group

	Net	Complementarity	Output (%)
Baseline (estimated η)	+56.9	+20.1	+0.71
$\eta = 0.70$	+70.9	+36.2	+0.70
$\eta = 0.76$ (pooled)	+64.9	+29.3	+0.71
$\eta = 1$	+39.7	0.0	+0.71

Baseline

- For every 100 jobs created in deregulated lines, **57** survive economy-wide
- **Complementarity** adds **20** jobs

Experiments: Complementarity Component per 100 jobs created in deregulated lines

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Net effect

- Baseline: **57** jobs survive economy-wide
- Homogeneous $\eta = 0.7 \rightarrow$ **25% too large**

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Net effect

- Homogeneous $\eta = 0.7 \rightarrow$ **25% too large**
- Pooled $\hat{\eta} = 0.76 \rightarrow$ **14% too large**

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$\eta = 1$	+39.7	0.0	+0.71

Net effect

- Pooled $\hat{\eta} = 0.76 \rightarrow$ **14% too large**
- Homogeneous $\eta = 1 \rightarrow$ **30% too small**
- The rest of manufacturing loses **60** jobs
- Output gain is the same in every row $\rightarrow$ output does not reveal jobs

Heterogeneity in η

A common η misattributes which sectors gain jobs per 100 jobs created in deregulated lines

By Group

	Complement sectors	Substitute sectors	Net
Baseline (estimated η)	+53.9	+3.0	+56.9
$\eta = 0.76$ (pooled)	+50.9	+14.0	+64.9
$\eta = 0.86$ (mean of $\hat{\eta}$)	+44.0	+10.7	+54.7

Sectors grouped by their *estimated* η in every row.

- Substitute sectors: about **one third** of employment
- Common η **quintuples** their job gains (14.0 vs 3.0)
- Complement sectors gain less (50.9 vs 53.9)
- Common η assigns **23** of every 100 jobs to the wrong sectors
- With $\eta = 0.86$ the aggregate is close (54.7 vs 56.9), the split is not

Which sectors does a reform reach? A complementarity index

- For each sector s: revenue-weighted, network-propagated sum of $(1-\eta)$ across every sector s' reached by s's price cuts
- The index accounts for
  - the size of the sectors reached ϑ + network position $\mathcal{L}$ + sectors reached technology.
- $\mathrm{Compl}(s) > 0 \longrightarrow$ price cuts reach mostly **complements**; < 0, mostly substitutes

$$\mathrm{Compl}(s) \;=\; \Big[\, \mathcal{L}\left(\vartheta \odot (\mathbf{1}-\eta)\right)\Big]_s, \qquad \mathcal{L} \;=\; \Big(\mathbf{I} - \mathbf{W}^{\top}\,\mathrm{diag}\left(\varphi_M(s)\right)\Big)^{-1}$$

- $W_{s's}$: share of buyer s''s materials spending on supplier s; $\varphi_M(s)$: materials share of revenue
- $\vartheta(s') = E(s')/\sum_{s''} E(s'')$: revenue weights

Two comparable reforms with different employment effects

	Reaches complements	Reaches substitutes
Comparable characteristics		
Employment deregulated (% of total)	0.71	0.71
Output gain (basis points)	4.2	4.3
Own η (average)	0.92	0.92
Average markup	1.22	1.23
Estimated technologies		
Effective η of the buyers reached	0.86	0.97
Net jobs, per 100 created	43.3	30.1
Labor–materials channel	22.3	3.2
Common elasticity, $\eta = 0.76$		
Net jobs, per 100 created	58.2	55.0
Labor–materials channel	38.0	30.1

- Two packages of 8 sectors, matched on output, employment and own η
- Only difference: the η of the buyers reached through the network

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Net jobs, per 100 created	58.2	55.0
Labor–materials channel	38.0	30.1

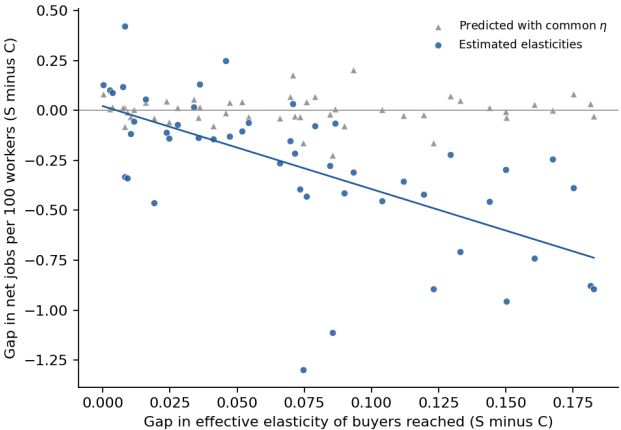
- Two packages of 8 sectors, matched on output, employment and own η
- Only difference: the η of the buyers reached through the network
- Estimated η: reaching complements creates **44%** more jobs

Two comparable reforms with different employment effects

	Reaches complements	Reaches substitutes
Comparable characteristics		
Employment deregulated (% of total)	0.71	0.71
Output gain (basis points)	4.2	4.3
Own η (average)	0.92	0.92
Average markup	1.22	1.23
Estimated technologies		
Effective η of the buyers reached	0.86	0.97
Net jobs, per 100 created	43.3	30.1
Labor–materials channel	22.3	3.2
Common elasticity, $\eta = 0.76$		
Net jobs, per 100 created	58.2	55.0
Labor–materials channel	38.0	30.1

- Two packages of 8 sectors, matched on output, employment and own η
- Only difference: the η of the buyers reached through the network
- Estimated η: reaching complements creates **44%** more jobs
- Common η: gap of 3 jobs (6%) $\rightarrow$ reforms look equivalent
- Relative to common η: **25%** fewer jobs (complements), **45%** fewer (substitutes)

Generalizing: 49 random pairs of reforms



- Pairs differ only in the effective η of the buyers reached
- Estimated η: buyers' η **0.25** higher $\rightarrow$ about **1** fewer net job per 100 workers in deregulated sectors
- Common η: job gaps around zero

Conclusions

- Pro-competition policy affects employment beyond the markets it directly deregulates.
- **Technology determines what happens to employment when shocks arrive.**
  - Cheaper materials raise labor demand in complementary sectors; reduce it in substitute.
- Complementarity accounts for about **one third** of the aggregate employment effect.
  - **83%** of its aggregate contribution arises outside directly deregulated sectors.
- Complementarity mechanism is more important than reduction in markups.
- **Heterogeneity** in η decides where jobs are created: comparable reforms differ by 44% in net jobs.
- Policy makers should consider not only I/O position but also technology of the **sectors reached.**

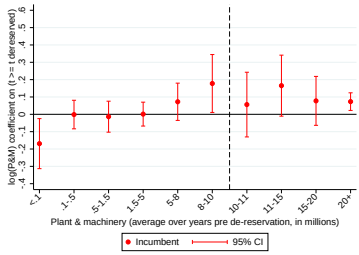
Appendix

Appendix: Calibration and Model Diagnostics

Incumbent Expansion

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- Pooled effect on capital of pre-policy binding firms.
  (Same specification as in Martin, Nataraj and Harrison, 2017)
- Δ log P&M of 0.178 (s.e. 0.085); model 0.071
  (Firms in the [8M-10M] bin)
- Relative to firms below Rs 8m: 0.200 in the data, 0.208 in the model (untargeted).
- In the model, we assume that firms in the [8M-10M] plant and machinery capital bin, are constrained by the cap.



Appendix: Parameter estimates

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Parameter	Source / target	Value	Dispersion
Panel A. Fixed or externally estimated			
γ within-market elasticity	set	6	—
θ_f across-market elasticity	set	1.25	—
κ supplier substitution elasticity	Peter & Ruane	2.5	—
α_K capital share	set	0.10	—
r rental rate	set	0.10	—
$\overline{B}$ statutory cap on P&M	policy	₹10m	—
$\eta(s)$ labor–materials elasticity	2-digit sector IV	med. 0.90	range [0.36, 1.73]
Panel B. Incumbents (per sector)			
$\xi(s)$ Pareto shape ($n = 238$)	U quantiles + HHI_U	4.30	IQR [3.16, 5.44]
$\sigma(s)$ R-type advantage	HHI_R, R quantiles	0.336	IQR [0.26, 0.41]
$\alpha(s)$ labor weight in bundle	factor shares	0.085	IQR [0.05, 0.13]
$\tau_i(s)$ capital conversion rate	eligible P&M	3.68	IQR [1.75, 7.30]
$K_i^*/\overline{K}_i$ cap severity	target 1.20	1.07	IQR [1.02, 1.39]
Panel C. Entrants (per sector)			
α_s^E entrant Pareto shape	entrant HHI; pooled elsewhere	3.42	IQR [3.29, 3.62]
φ_s entrant location	entrant sales share	0.86	IQR [0.74, 0.99]
$N^E(s)$ entrants per sector	data (four-period median)	5	IQR [1, 12]

Appendix: Operating and entry costs

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	What it equals	Median	IQR
Identified by revealed preference			
$f^R(s)$	gross profit of the marginal R incumbent	1.1×10^{-6}	$[2.5 \times 10^{-7},\ 1.5 \times 10^{-5}]$
$f^U(s)$	gross profit of the marginal U incumbent	9.7×10^{-5}	$[2.8 \times 10^{-5},\ 4.9 \times 10^{-4}]$
$f^E(s)$	midpoint of $[\underline{f}^E,\ \bar{f}^E]$ ($n = 83$)	1.9×10^{-4}	$[7.4 \times 10^{-5},\ 6.6 \times 10^{-4}]$
Unit-free ratios			
f^E/f^R	sunk cost vs marginal operating cost	302	[46, 1,517]
f^U/f^R	never-reserved vs reserved operating cost	≈ 90	ratio of medians

Appendix: Untargeted moments

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Moment	*N*	Data	Model
Incumbents with Rs 8–10m before the reform: plant and machinery			
Δ log P&M	120	0.178 (0.085)	0.071
relative to incumbents below Rs 8m		0.200	0.208
Share increasing P&M	120	0.649	0.620
Share above Rs 10m after the reform	120	0.558	0.613
Share remaining in Rs 8–10m	120	0.157	0.268
Incumbents with Rs 8–10m before the reform: market shares			
Median ratio of post- to pre-reform share	120	0.586	0.931
Share losing market share	120	0.716	0.694
Concentration of reserved markets, by sector			
Mean change in HHI	94	−0.062	−0.029
Median change in HHI	94	−0.013	−0.007
Share of sectors where HHI falls	94	0.543	0.755
Entrant P&M p50 / cap, if above the cap	83	4.15	5.43

Entrants

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Where entrant productivity comes from

The number of entrants $N^E(s)$ is the observed count (median over T to $T+3$). Entrants draw productivity from a Pareto distribution anchored to incumbents, on a quantile grid:

$$z_{js}^E = \bar{z}_s^R \, \varphi_s \, (1 - u_{js})^{-\frac{1}{\alpha_s^E}}, \qquad u_{js} = \frac{j - \frac{1}{2}}{N^E(s)}, \quad j = 1, \ldots, N^E(s)$$

- $\bar{z}_s^R$: mean productivity of the sector's reserved incumbents
- Location $\varphi_s \to$ entrant sales share
- Tail $\alpha_s^E \to$ entrant sales HHI (75 sectors; pooled 3.42 elsewhere)

Removing Cap $\overline{K}$ - Entry and Expansion Margins

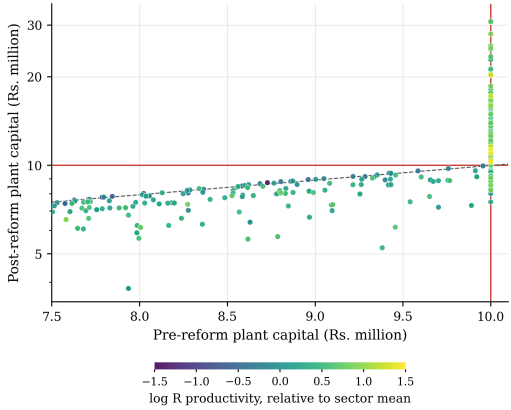
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	Model	
Entrants operating	1,210	of 1,223 in the cohort
Reserved sectors with entry	104	of 113 reserved sectors
Eligible incumbents increasing P&M	429	4.1% of 10,443
of which ending above Rs 10m	427	of 519 at the cap before
Reserved sectors with incumbent expansion	57	of 108 with eligible incumbents

- Non-negligible intensive and extensive entry margin.
- Entrants per sector: median 5 (IQR 1–12).

Removing Cap $\overline{K}$ - Expansion Diagnosis

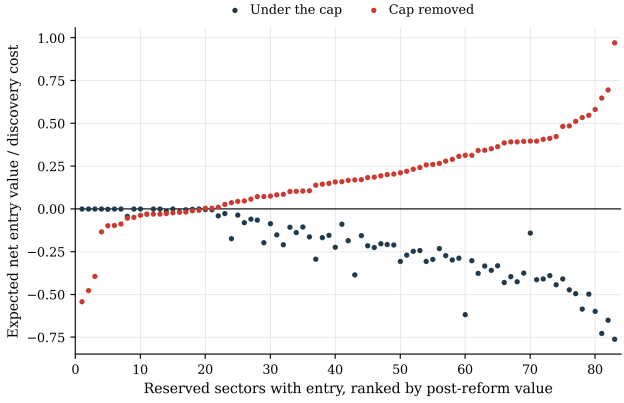
Back



- Model generates expansion in a set of previously capped firms.
- Interaction with entry generates some firms in the cap to reduce the capital holdings

Removing Cap $\overline{K}$ - Net Profits Potential Entrants Diagnosis

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- Unfeasible entry in the baseline
- $\overline{K}$ removal $\longrightarrow$ set of potential entrants overcome sunk costs and enter

Appendix: Identification

Identification of Sunk Cost and operation costs

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- Entrants pay the sunk cost $f^E(s)$ *before* learning their productivity $\rightarrow$ compare it with the expected option value
- Lower bound: expected value of entering **under the cap**, alone, at pre-reform prices

$$\underline{f}^E(s) = \frac{1}{N^E(s)} \sum_j \max\left\{\Pi_j^{cc} - f^R(s),\, 0\right\}$$

- Upper bound $\overline{f}^E(s)$: same, **without the cap**, with the whole cohort (Bertrand equilibrium)
- No entry under the cap + entry after its removal $\rightarrow \underline{f}^E(s) < f^E(s) \leq \overline{f}^E(s)$
- $f^E(s)$ is the midpoint of the interval (non-empty in 89 of 113 sectors)
- Given the observed entrant count, $f^E(s)$ is sunk at the operating stage: it does not move the counterfactual

Identifying cap severity from MRPK

Pair Reserved and Never Reserved branches of the same 3-digit sector

- Follow Hsieh and Klenow (2009) logic:
  - Uncapped case: firms optimize at $MRPK = r$. **Never-reserved firms**.
  - Capped case, we have $MRPK(\overline{K})$. **Reserved firms**.
  - Only consider firms with $K \in [8M \;\; 10M]$

$$r = MRPK(\overline{K}) \times \left(\frac{K^*}{\overline{K}}\right)^{\frac{-1}{1-\alpha}} \quad \Longrightarrow \quad \frac{K^*}{\overline{K}} = \left(\frac{\mathrm{MRPK}(\overline{K})}{r}\right)^{1-\alpha}$$

Measurement

$$\widehat{\tau}_s = \left(\frac{\widehat{r}}{MRPK(\overline{K})}\right)_s = \frac{\mathrm{med}\{\mathrm{MRPK}_i : \; i \in \mathcal{R}_s, \; K_i \in B\}}{\mathrm{med}\{\mathrm{MRPK}_j : \; j \in \mathcal{U}_s, \; K_j \in B\}}, \qquad \hat{\tau} = \mathrm{med}_s \; \widehat{\tau}_s$$

Model targets 1.20; median cap severity $K_i^*/\overline{K}_i$ in the model 1.07 (IQR 1.02–1.39)

Identification of sunk and operation costs

Sunk Costs | Operating Costs

- Only entrants pay sunk costs
- Entrants and Incumbents pay operation costs.
- Sunk cost is such that potential entrants would not enter under the cap, but enter once it is removed.
- Operation costs are identified in the incumbent-only baseline.
- Minimum cost such that every observed incumbent in R and U stay in the market

Identification of Operational Costs

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- Largest cost consistent with every observed R incumbent continuing

- Largest cost consistent with every observed U incumbent continuing

- In the baseline, I assume that incumbents already paid the one-shot sunk cost

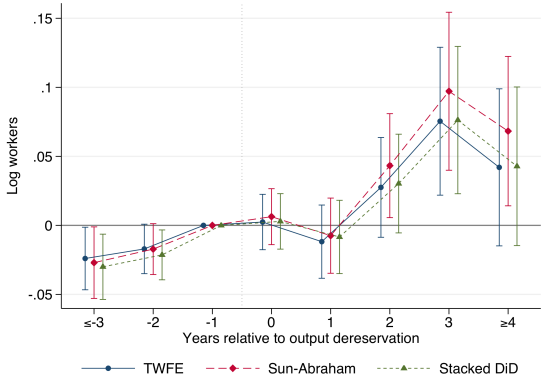
Appendix: Empirical Results

Output dereservation $\longrightarrow \downarrow$ Markup $+ \uparrow$ Labor

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Effect of output dereservation on log markups

	(1)	(2)	(3)
Post output dereservation	−0.096**	−0.103***	−0.139***
	(0.038)	(0.038)	(0.051)
Product FE	Yes	Yes	Yes
NIC2 × year FE	Yes	Yes	Yes
State × year FE	No	Yes	No
Firm FE	No	No	Yes
Pre-trends p-value	0.149	0.174	0.419
Observations	214,294	214,294	187,611



$$\log \mu_{ipt} = \beta^{O} \, Post^{O}_{ipt} + \sum_{k \in \mathcal{K}^{pre}} \gamma^{O}_{k} \, \mathbf{1}\{\tau^{O}_{ipt} = k\} + \alpha_p + \delta_t + \lambda_{s \times t} + \varepsilon_{ipt}$$

$$\log L_{it} = \sum_{k \in \mathcal{K}} \beta^{O}_{k} \, \mathbf{1}\{\tau^{O}_{it} = k\} + \alpha_i + \delta_t + \varepsilon_{it}$$

Appendix: Decomposition and Experiments

Decomposition of the Industrial Policy Effects on Prices

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		Expansion	Entry	Total
All sectors	Output price	−0.04	−0.66	−0.69
	Materials price	−0.02	−0.43	−0.45
	Markup	0.00	−0.03	−0.03
Reserved	Output price	−0.34	−5.24	−5.52
	Materials price	−0.04	−0.75	−0.78
	Markup	−0.01	−0.48	−0.48
Never-reserved	Output price	−0.01	−0.34	−0.35
	Materials price	−0.02	−0.41	−0.42
	Markup	0.00	0.00	0.00

Percentage changes, baseline revenue weights.

- Entry is the dominant mechanism
- Reserved sectors:
  - ↓ 5.5% Output price
  - ↓ 0.8% Materials price
- Never Reserved sectors – Network propagation:
  - ↓ 0.42% Materials price
  - ↓ 0.35% Output price
- Productivity in deregulated lines ↑ 4.1 log points

Experiments: Complementarity Component by Group

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Net jobs per 100 created in deregulated lines

	Deregulated		Never-reserved		
	$\eta < 1$	$\eta > 1$	$\eta < 1$	$\eta > 1$	Net
Baseline (estimated η)	+80.8	+19.2	−26.9	−16.2	+56.9
$\eta = 0.70$	+80.1	+19.9	−25.1	−4.0	+70.9
$\eta = 0.76$	+80.1	+19.9	−29.2	−5.9	+64.9
$\eta = 0.86$ (mean of $\hat{\eta}$)	+80.0	+20.0	−36.1	−9.3	+54.7
$\eta = 1$	+80.0	+20.0	−46.1	−14.2	+39.7

Groups use the estimated η in every row; each row scaled by its own deregulated jobs.

- η changes the aggregate almost entirely through what happens outside the deregulated sectors.
- Common η: never-reserved substitute sectors lose 4–9 jobs instead of 16.
- Assumptions on η meaningfully change the results!

Decomposition of the Industrial Policy Effects on Employment: Within group effect

Change in employment, % of the group's baseline

	Scale	Markup	Compl.	Total
Deregulated	+3.27	+0.38	+0.13	+3.79
Never-reserved	−0.23	0.00	+0.07	−0.17
Economy	+0.09	+0.04	+0.07	+0.20

- Dereserved sectors ↑ employment 3.8%
  Represent 6.5% of manufacturing revenue, 9.4% of employment
- Never-regulated sectors ↓ employment 0.17%
- Complementarity acts as a 'cushion' mechanism in never affected sectors.
  Offsets 28% of the scale losses
- Total aggregate = +0.20% of agg. empl.
  Large gain on a few sectors, small loss on a large base